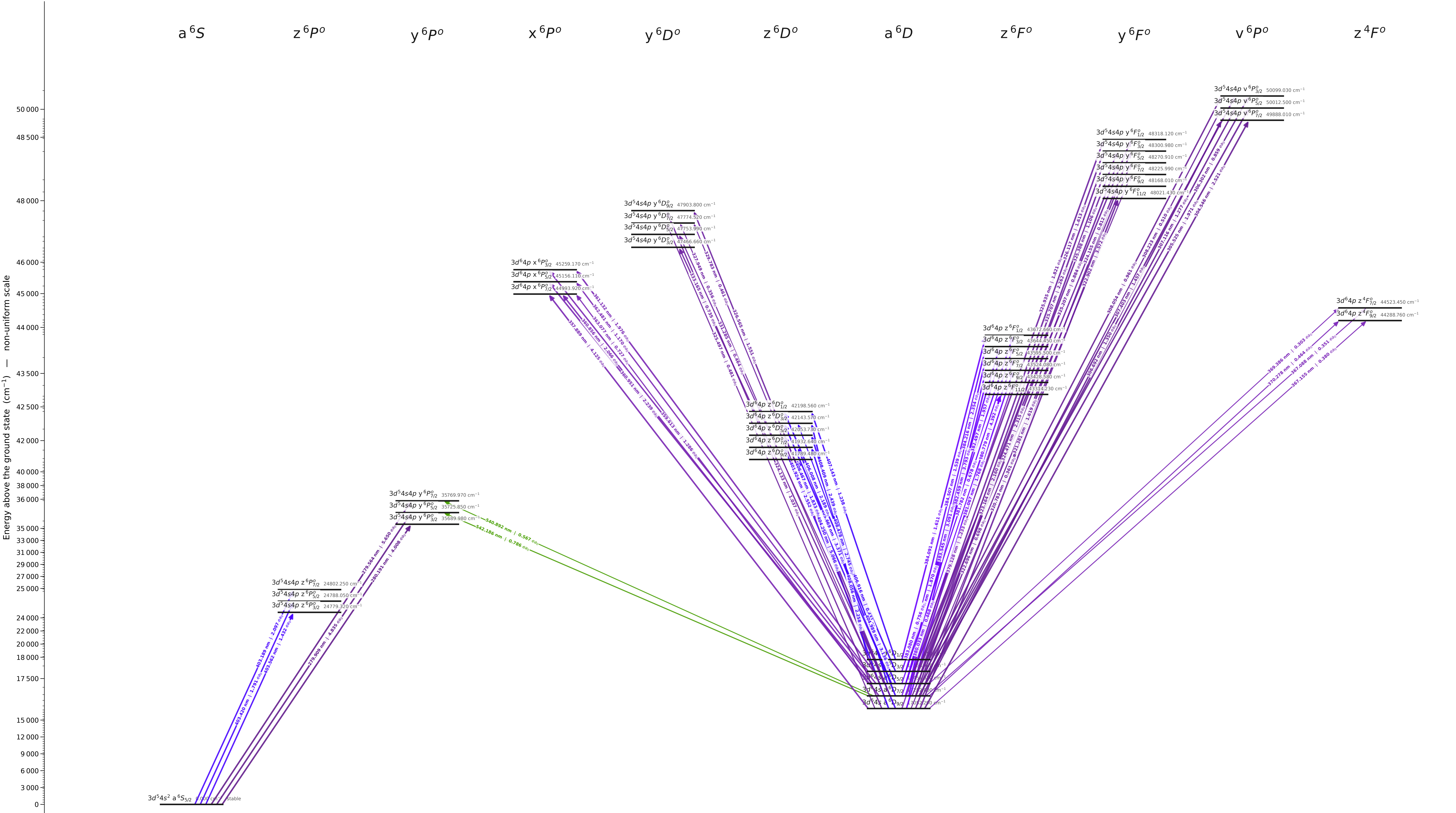
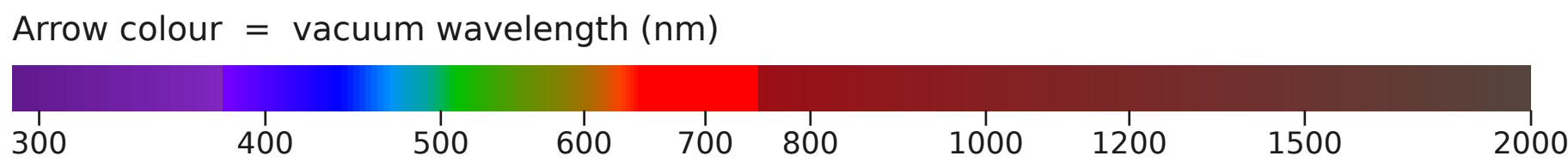


Manganese

Mn (⁵⁵Mn I). Strongest classified transitions below 2 μm (NIST): 41 levels, 70 lines.

Mn⁺ ionisation limit 59959.56 cm⁻¹ = 7.43404 eV (166.779 nm)

Arrow label: reduced dipole matrix element $|\langle J||er||J'\rangle|$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J er J'\rangle $ (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3d ⁵ 4s ² a ⁶ S _{5/2} - 3d ⁵ 4s4p y ⁶ P _{3/2} ^o	279.5641	279.4817	1072.3567	5.650	3.700e+08	-	-	
3d ⁵ 4s ² a ⁶ S _{5/2} - 3d ⁵ 4s4p y ⁶ P _{5/2} ^o	279.9094	279.8269	1071.0340	4.835	3.600e+08	-	-	
3d ⁵ 4s ² a ⁶ S _{5/2} - 3d ⁵ 4s4p y ⁶ P _{1/2} ^o	280.1907	280.1081	1069.9587	4.008	3.700e+08	-	-	
3d ⁶ 4s a ⁶ D _{9/2} - 3d ⁵ 4s4p y ⁶ P _{3/2} ^o	304.5464	304.4578	984.3901	2.521	5.700e+07	-	-	
3d ⁶ 4s a ⁶ D _{7/2} - 3d ⁵ 4s4p v ⁶ P _{5/2} ^o	305.5254	305.4366	981.2357	1.971	4.600e+07	-	-	
3d ⁶ 4s a ⁶ D _{9/2} - 3d ⁵ 4s4p y ⁶ F _{11/2} ^o	322.9021	322.8089	928.4315	3.572	6.400e+07	-	-	
3d ⁶ 4s a ⁶ D _{5/2} - 3d ⁵ 4s4p y ⁶ F _{5/2} ^o	324.4711	324.3775	923.9421	2.315	5.300e+07	-	-	
3d ⁶ 4s a ⁶ D _{3/2} - 3d ⁵ 4s4p y ⁶ F _{3/2} ^o	325.7071	325.6132	920.4357	2.262	5.000e+07	-	-	
3d ⁶ 4s a ⁶ D _{1/2} - 3d ⁵ 4s4p y ⁶ F _{1/2} ^o	325.9349	325.8410	919.7923	1.821	9.700e+07	-	-	
3d ⁶ 4s a ⁶ D _{9/2} - 3d ⁶ 4p x ⁶ P _{5/2} ^o	357.8889	357.7868	837.6690	4.125	9.400e+07	-	-	
3d ⁶ 4s a ⁶ D _{9/2} - 3d ⁶ 4p z ⁶ F _{11/2} ^o	380.7792	380.6711	787.3131	4.392	5.900e+07	-	-	
3d ⁶ 4s a ⁶ D _{7/2} - 3d ⁶ 4p z ⁶ F _{9/2} ^o	382.4592	382.3507	783.8547	3.793	5.210e+07	-	-	
3d ⁶ 4s a ⁶ D _{1/2} - 3d ⁶ 4p z ⁶ F _{1/2} ^o	384.0908	383.9819	780.5249	1.611	4.640e+07	-	-	
3d ⁶ 4s a ⁶ D _{9/2} - 3d ⁶ 4p z ⁶ D _{9/2} ^o	404.2496	404.1355	741.6023	5.066	7.870e+07	-	-	
3d ⁶ 4s a ⁶ D _{5/2} - 3d ⁶ 4p z ⁶ D _{5/2} ^o	404.9887	404.8743	740.2490	3.136	7.500e+07	-	-	
3d ⁶ 4s a ⁶ D _{3/2} - 3d ⁶ 4p z ⁶ D _{3/2} ^o	406.0076	405.8930	738.3912	2.189	7.250e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).